National University of Computer and Emerging Sciences

Lahore Campus

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Analogue & Digital Communication (EE3010/3003)

Date: November 2, 2026

Course Instructor(s)

1. Dr. S. M. Sajid (CM) (BEE-5A/7A, 5C/7C)
2. Mohsin Yousuf (BEE-5B/7B, 5D/7D)

Sessional-II Exam

Total Time (Hrs):	**1**
Total Marks:	**60**
Total Questions:	**3**

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Roll No Section Student Signature

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1. **Attempt all questions and remember to solve parts of the same question together.**
2. **Show all the steps with the help of diagrams and equations.**

***CLO # 02: Calculate** parameters like bandwidth, power efficiency, transmission efficiency, and modulation index for linear (AM, SSB, VSB, QAM) and non-linear (FM, PM) analog modulation systems. [C3]*

Q1: A multi-tone message signal given by, **[20 marks]**

$$m(t) = 4\cos(2\pi \times 10^3 t) + 2\cos(2\pi \times 3 \times 10^3 t)$$

amplitude modulates a carrier wave, $c(t) = 10\cos(2\pi \times 10^6 t)$ to produce a conventional AM (DSB-With Carrier) signal.

(a) **Determine** the peak amplitude of the message signal and the resulting modulation index μ. **[5]**

(b) **Calculate** the total transmission power P_t and the power transmission efficiency η of this AM system. **[10]**

(c) **Find** the transmission bandwidth B_{AM} required to broadcast this signal without crosstalk. **[5]**

Q2: An angle modulator with a frequency deviation sensitivity of $k_f = 2\pi \times 10^4$ [rad/s/V] is excited by a single-tone message signal $m(t) = 3\cos(2\pi \times 2 \times 10^3 t)$ V to produce a Frequency Modulated (FM) wave. **[20 marks]**

(a) **Calculate** the peak frequency deviation Δf in Hz and the corresponding FM modulation index β. **[6]**

(b) **Estimate** the essential transmission bandwidth B_{FM} of the modulated signal using WBFM approximation (Carson's Rule). **[6]**

(c) If the amplitude of the message signal is doubled while its frequency is halved, **determine** the new frequency deviation Δf_{new} and **calculate** the new WBFM bandwidth. **[8]**

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***CLO # 03: Apply** the sampling theorem and analog-to-digital pulse conversion techniques (PAM, PWM, PPM, PCM, DPCM, DM) for baseband processing. [C3]*

Q3: An analogue signal $m(t) = 5\cos(1000\pi t) + 3\cos(4000\pi t)$ is to be converted into a *binary* digital signal using a uniform Pulse Code Modulation (PCM) system. The signal is sampled at a rate 25% higher than its Nyquist rate. The dynamic range of the quantizer matches the peak-to-peak amplitude of $m(t)$, and the maximum allowable quantization error must not exceed 0.5% of the peak amplitude of the signal. **[20 marks]**

(a) **Determine** the Nyquist rate of the signal f_N and **find** the actual sampling frequency f_S. **[6]**

(b) **Calculate** the minimum number of bits per sample (n) required to satisfy the quantization error constraint, and **state** the actual number of uniform levels (L) used. **[6]**

Hint: For uniform quantization with step size $\Delta = \frac{2m_p}{L}$, the maximum error is:

$$|\epsilon|_{max} = \frac{\Delta}{2} = \frac{m_p}{L}, \qquad \text{where } L = 2^n \text{ for binary PCM}$$

(c) **Compute** the resulting minimum bit rate (R_b) of the PCM line stream and find the minimum absolute channel bandwidth needed for distortionless baseband transmission. If T1 carrier system is utilized using TDM, i.e. 24 such signals are time-multiplexed, again **determine** the minimum transmission bandwidth required to transmit the multiplexed signal. **[8]**

- End of Question Paper -

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